Basic C++ Program breakdown (class-object-method)

Class

```
class ClassName {
private:
    // Data members (attributes/variables)
    // Only accessible within class

public:
    // Member functions (methods)
    // Accessible from outside class

    // Constructor
   (special method - runs when object is created)
    internal ClassName() {
        // Initialization code
    }

    // Regular methods
    returnType methodName(parameters) {
        // Method body
    }
};
```

Application

```
#include <iostream>
#include <string>
using namespace std;

class Car {
private:
    string brand;
    string model;
    int year;

public:
    // Method to set car details
    void setDetails(string b, string m, int y) {
        brand = b;
        model = m;
        year = y;
    }

    // Method to display car details
    void displayDetails() {
        cout << "Car Details:" << endl;
        cout << "Brand: " << brand << endl;
        cout << "Model: " << model << endl;
        cout << "Year: " << year << endl;
    }

    // Method to check if car is modern
    bool isModern() {
        // Returns true if year is after 2015
        if (year > 2015) {
            return true;
        } else {
            return false;
        }
    }
};

int main() {
    // Create two car objects
    Car car1, car2;

    // Set details for car1
    car1.setDetails("Toyota", "Corolla", 2018);

    // Set details for car2
    car2.setDetails("Honda", "Civic", 2010);

    // Display car1 details
    cout << "=== Car 1 ===" << endl;
    car1.displayDetails();
    if (car1.isModern()) {
        cout << "This is a modern car." << endl;
    } else {
        cout << "This is not a modern car." << endl;
    }

    cout << endl;

    // Display car2 details
    cout << "=== Car 2 ===" << endl;
    car2.displayDetails();
    if (car2.isModern()) {
        cout << "This is a modern car." << endl;
    } else {
        cout << "This is not a modern car." << endl;
    }

    return 0;
}
```

Spotlight

```
#include <iostream>
#include <string>
using namespace std;

class Car {
private:
    string brand;
    string model;
    int year;

public:
    // Method to set car details
    void setDetails(string b, string m, int y) {
        brand = b;
        model = m;
        year = y;
    }

    // Method to display car details
    void displayDetails() {
        cout << "Car Details:" << endl;
        cout << "Brand: " << brand << endl;
        cout << "Model: " << model << endl;
        cout << "Year: " << year << endl;
    }

    // Method to check if car is modern
    bool isModern() {
        // Returns true if year is after 2015
        if (year > 2015) {
            return true;
        } else {
            return false;
        }
    }
};

int main() {
    // Create two car objects
    Car car1, car2;

    // Set details for car1
    car1.setDetails("Toyota", "Corolla", 2018);

    // Set details for car2
    car2.setDetails("Honda", "Civic", 2010);

    // Display car1 details
    cout << "=== Car 1 ===" << endl;
    car1.displayDetails();
    if (car1.isModern()) {
        cout << "This is a modern car." << endl;
    } else {
        cout << "This is not a modern car." << endl;
    }

    cout << endl;

    // Display car2 details
    cout << "=== Car 2 ===" << endl;
    car2.displayDetails();
    if (car2.isModern()) {
        cout << "This is a modern car." << endl;
    } else {
        cout << "This is not a modern car." << endl;
    }

    return 0;
}
```